

Visual gate detection :

Worst cases

Case 1:



Gates overlap

↳ go with biggest gate area candidate first

Case 2:



Two gates of approximately same area

↳ Pick one with right handmost centre

i.e go for 1) Closest gate

2) Right handmost

3) Maths: possibility or need to "complete" partial gates?

Case 3:



Partial gate front

↳ just go forward, with safety padding?



Partial 2nd gate

↳ just aim for first one. Note: can add

sanity check : if no gate detected in gate zone
-> fallback gate search procedure
e.g. revert to previous grey zone, perform visual sweep

Path Planning Worst Cases

Test process:

- Go through cases 1 to 5 with remote control, max fov
- test and comment on proposed strategies.

- test gate visibility / distributions with max fov
- Test accuracy of drone perceived position in arena.



